SCHIZOPHRENIA

Schizophrenia is one of the most complex and challenging psychiatric disorders as it represents a heterogeneous syndrome of disorganized and bizarre thoughts, delusions, hallucinations, inappropriate affect, and impaired psychosocial functioning.

ETIOLOGY

Although multiple neurotransmitter dysfunctions are involved in schizophrenia, the etiology is more likely mediated by multiple subcellular processes that are influenced by different genetic polymorphisms.

First-degree relatives of patients with schizophrenia carry a 10% risk of developing the disorder. When both parents have the diagnosis, the risk to their offspring is 40%.

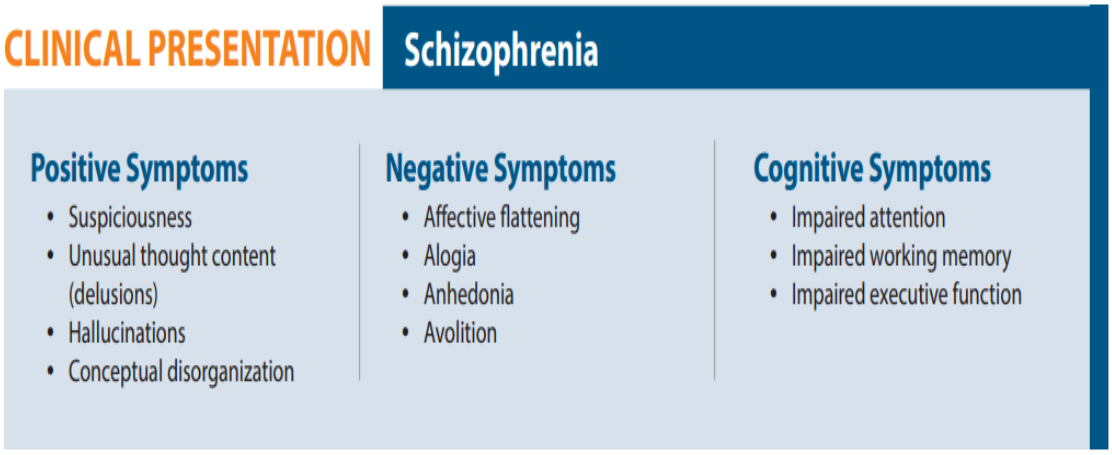
Biological factors : ► Genetic factors ► Biochemical abnormalities ► Abnormal brain structures ► Viral problems

Psychological factors : ▪ Psychodynamic ▪ Cognitive

Socio-culture factor: ▪ Social labelling ▪ Family dysfunctioning

PATHOPHYSIOLOGY (FOR UNDERSTANDING)

- Increased ventricular size and decreased gray matter, have been reported.
- Schizophrenia causation theories include genetic predisposition, obstetric complications, increased neuronal pruning, immune system abnormalities, neurodevelopmental disorders, neurodegenerative theories, dopamine receptor defect, and regional brain abnormalities including hyper- or hypo-activity of dopaminergic processes in specific brain regions.
- Positive symptoms may be more closely associated with dopamine receptor hyperactivity in the mesocaudate, whereas negative and cognitive symptoms may be most closely related to dopamine receptor hypofunction in the prefrontal cortex.
- Glutamatergic dysfunction. A deficiency of glutamatergic activity produces symptoms similar to those of dopaminergic hyperactivity and possibly schizophrenic symptoms.
- Serotonin (5-hydroxytriptamine [5-HT]) abnormalities. Schizophrenic patients with abnormal brain scans have higher whole blood 5-HT concentrations, which correlate with increased ventricular size



Hallucination (loss of touch from reality) – 5 types

- ✔ Auditory hallucinations
- ✔ Visual
- ✔ Smell
- ✔ Taste
- ✔ Touch

Delusion- it is a false belief

Types-

1. Delusion of reference- refers to everything going around them
2. Paranoid delusion- this delusion reflects profound fear along with the loss of ability to tell them what is real and what is not. They might make you feel like people are trying to hurt you or poison your food. It also may suggest that your spouse or partner is cheating on you.
3. Delusion of grandeur- extra ability of something

MEDICATIONS/ILLICIT DRUGS THAT CAN CAUSE PSYCHOTIC SYMPTOMS

- ✔ Anticholinergics (centrally-acting, high doses)
- ✔ Cannabis
- ✔ Dextromethorphan
- ✔ Dopamine or dopamine agonists (e.g.. Requip, Mirapex,Sinemet)
- ✔ Interferons
- ✔ Stimulants, especially if already at risk (includes amphetamines)
- ✔ Systemic steroids (typically with lack of sleep -ICU psychosis)

Illicit substances

- ✔ Bath salts (synthetic cathinones, MDPV)
- ✔ Cocaine, especially "crack" cocaine
- ✔ Lysergic acid diethylamide (LSD,hallucinogenics)
- ✔ Methamphetamine, ice, crystal
- ✔ Phencyclidine (PCP)

DIAGNOSIS

DSM-5 DIAGNOSTIC CRITERIA FOR SCHIZOPHRENIA

Note: delusions, hallucinations or disorganized speech must be present

Negative signs and symptoms	Positive signs and symptoms
Loss of interest in everyday activities	Hallucinations: can be auditory (hearing voices), visual, or somatic
Lack of emotion (apathy)	Delusions: beliefs held by the patient that are without a basis in reality
Inability to plan or carry out activities	Disorganized thinking/behavior, incoherent speech, often on unrelated topics, purposeless behavior, or difficulty speaking and organizing thoughts, such as stopping in mid-sentence or jumbling together meaningless words
Poor hygiene	Difficulty paying attention
Social withdrawal	
Loss of motivation (avolition)	
Lack of speech (alogia)	

TABLE 76-2 DSM-IV-TR Diagnostic Criteria for Schizophrenia

A. Characteristic symptoms: Two or more of the following, each persisting for a significant portion of at least a 1-month period:
   (1) Delusions
   (2) Hallucinations
   (3) Disorganized speech
   (4) Grossly disorganized or catatonic behavior
   (5) Negative symptoms

Note: Only one criterion A symptom is required if delusions are bizarre or if hallucinations consist of a voice keeping a running commentary on the person's behavior or two or more voices conversing with each other.

B. Social/occupational dysfunction: For a significant portion of the time since onset of the disorder, one or more major areas of functioning such as work, interpersonal relations, or self-care are significantly below the level prior to onset.

C. Duration: Continuous signs of the disorder for at least 6 months. This must include at least 1 month of symptoms fulfilling criterion A (unless successfully treated). This 6 months may include prodromal or residual symptoms.

D. Schizoaffective or mood disorder has been excluded.

E. Disorder is not due to a medical disorder or substance use.

F. If a history of a pervasive developmental disorder is present, there must be symptoms of hallucinations or delusions present for at least 1 month.

DSM-IV-TR, Diagnostic and Statistical Manual of Mental Disorders, 4th ed., text revision.
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TYPES OF SCHRIZOPHRENIA

Catatonic : a little to no movement

Disorganised : a common achy type of schizophrenia, ▪ Disorganised thinking ▪ Flat effect ▪ Inappropriate emotions or behaviour

Paranoid : common achy type of schizophrenia; delusion, hallucinations, false belief

Residual : Long term schizophrenia with most symptoms have disappeared, negative symptoms(detraction from normal behaviour) often remain such as a flat effect or refusal to talk.

Undifferentiated : Does not fit in one of the above categories because the patient suffers form symptoms of multiple types.

TREATMENT-

NON-PHARMACOLOGICAL-

Psychosocial rehabilitation programs oriented toward improving patients' adaptive functioning are the mainstay of nondrug treatment for schizophrenia. These programs can include case management, psychoeducation, targeted cognitive therapy, basic living skills, social skills training, basic education, work programs supported housing, and financial support. In particular, programs aimed at supportive employment and housing are effective and considered "best practices".

PHARMACOTHERAPY-

FIRST-GENERATION ANTIPSYCHOTICS (TYPICAL)

First-generation antipsychotics (FGAs) work mainly by blocking dopamine (D2) receptors, and minimal serotonin (5HT2A) receptor blockade.

DRUG	DOSING	SAFETY/SIDE EFFECTS/MONITORING
Low Potency		**BOXED WARNINGS** Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo Thioridazine: QT prolongation *Adasuve*: bronchospasm (REMS program) **SIDE EFFECTS** Sedation and EPS: lower potency drugs have ↑ sedation and ↓ EPS, and higher potency drugs have ↓ sedation with ↑ EPS Leukopenia, neutropenia, and agranulocytosis (sometimes fatal) Seizures (phenothiazines, butyrophenones), temperature dysregulation, allergic reactions; hepatic, ophthalmologic and hematologic side effects Cardiovascular effects: orthostasis, tachycardia (IV haloperidol has high risk) QT prolongation (especially with thioridazine, haloperidol, chlorpromazine) Sexual dysfunction *Adasuve*: dysgeusia (bad, bitter, or metallic taste in mouth), sedation
Chlorpromazine	300-1,000 mg/day, divided	
Thioridazine	300-800 mg/day, divided	
Mid potency		
Loxapine (*Loxitane*, *Adasuve*-inhalation powder for acute agitation)	30-100 mg/day, divided	
Perphenazine	8-64 mg/day, divided	
High Potency		
Haloperidol (*Haldol*, *Haldol Decanoate*) Oral, IV, Decanoate IM Class: butyrophenone Also used for Tourette syndrome (for tics and vocal outbursts)	Oral (tablet, solution): start 0.5-2 mg BID-TID, up to 30 mg/day IV: usually 5-10 mg Decanoate (monthly): IM only; for conversion from PO, use 10-20x the PO dose	
Fluphenazine Tablet, elixir, injectable, IM decanoate	6-12 mg/day, divided Decanoate (every 2 weeks): IM only	
Thiothixene	15-60 mg/day, divided	
Trifluoperazine	15-50 mg/day, divided	

- High-potency FGAs, such as haloperidol, are associated with a high risk of EPS effects, a moderate risk of sedation and a lower risk of orthostatic hypotension, tachycardia and anticholinergic effects compared to low-potency FGAs.
- low-potency FGAs are associated with a lower risk of EPS, a high degree of sedation, and a high risk of cardiovascular effects (orthostatic hypotension, tachycardia) and anticholinergic effects.
- Other side effects vary by the specific medication. FGAs are associated with a moderate risk of weight gain, a low risk of metabolic effects, a higher risk of EPS and a risk of sexual side effects.

SECOND-GENERATION ANTIPSYCHOTICS (ATYPICAL)

Second-generation antipsychotics (SGAs) work by blocking dopamine (D2) and serotonin (5HT2A) receptors.

Aripiprazole, brexpiprazole and cariprazine are unique: they act as D2 and 5HT1A partial agonists. Brexpiprazole also acts as a 5HT2A antagonist

DRUG	DOSING	SAFETY/SIDE EFFECTS/MONITORING
Aripiprazole (*Abilify*, *Abilify Maintena*, *Aristada* injection) Tablet, ODT, IM suspension Also approved for irritability with autism and Tourette disorder	10-30 mg PO QAM *Abilify Maintena*-IM suspension, give monthly *Aristada*-IM suspension, give every 4-8 weeks	**SIDE EFFECTS** Akathisia, anxiety, insomnia, constipation Less weight gain, some QT prolongation **NOTES** *Aristada* frequency dependent on dose
Clozapine (*Clozaril*, *FazaClo ODT*, *Versacloz* suspension) Only if failed to respond to treatment with 2 standard AP treatments, or had significant ADRs	300-900 mg/day, divided (start at 12.5 mg and titrate, also titrate off since abrupt discontinuation can cause seizures) Clozapine is very effective and has ↓ risk of EPS/TD, but used no sooner than 3rd line due to severe side effect potential (metabolic effects, neutropenia)	**BOXED WARNINGS, CLOZAPINE-SPECIFIC** Significant risk of potentially life-threatening neutropenia/agranulocytosis (REMS program) Bradycardia, orthostatic hypotension, syncope, and cardiac arrest; risk is highest during the initial titration period especially with rapid dose increases – titrate slowly Myocarditis and cardiomyopathy; discontinue if suspected Seizures, dose related; start at no higher than 12.5 mg once or twice daily, titrate slowly, using divided doses; use with caution in patients at seizure risk (e.g., seizure history, head trauma, alcoholism or taking medications which lower seizure threshold) **SIDE EFFECTS** Orthostasis, syncope, weight gain, ↑ lipids, ↑ glucose, somnolence, dizziness, insomnia, GI upset, sialorrhea (hypersalivation), QT prolongation, agranulocytosis, seizures, myocarditis **MONITORING** REMS: prescribers and pharmacies must be certified and patients must enroll with the Clozapine REMS To start treatment, baseline ANC must be ≥ 1,500/mm³ Check ANC weekly x 6 months, then every 2 weeks x 6 months, then monthly Stop therapy if ANC < 1,000/mm³ **NOTES** Smoking reduces drug levels
Lurasidone (*Latuda*)	40-160 mg/day, divided Take with food ≥ 350 kcal	**CONTRAINDICATIONS** Use with strong CYP450 3A4 inducers and inhibitors **SIDE EFFECTS** EPS, somnolence, dystonias, nausea, agitation, akathisia Nearly weight, lipid and blood glucose neutral
Olanzapine (*Zyprexa*, *Zydis ODT*, *Zyprexa Relprevv* injection) + fluoxetine (*Symbyax*) for treatment-resistant depression	10-20 mg QHS IM Injection (acute agitation) *Relprevv* inj suspension lasts 2-4 weeks; restricted use	**BOXED WARNING** *Zyprexa Relprevv* – sedation (including coma) and delirium (including agitation, anxiety, confusion, disorientation) have been observed following injection; must be administered in a registered healthcare facility, and patients must be monitored for 3 hours post-injection (REMS program requirements). **SIDE EFFECTS** Somnolence, weight gain, ↑ lipids, ↑ glucose, EPS, QT prolongation (lower risk), multiorgan hypersensitivity (DRESS) reactions **NOTES** Smoking reduces drug levels

DRUG	DOSING	SAFETY/SIDE EFFECTS/MONITORING
Paliperidone *(Invega, Invega Sustenna, Invega Trinza)* Active metabolite of risperidone; SEs similar	PO: 3-12 mg daily CrCl < 50 mL/min: 3 mg daily CrCl < 10 mL/min: Not recommended OROS delivery enables once daily dosing-do not break or crush *Invega Sustenna*, IM injection, give monthly *Invega Trinza*, IM injection, give every 3 months (start only after receiving *Invega Sustenna* x 4 months)	**SIDE EFFECTS** ↑ prolactin – sexual dysfunction, galactorrhea, irregular/missed periods EPS, especially at higher doses Tachycardia, headache, sedation, anxiety QT prolongation Weight gain, ↑ lipids, ↑ glucose
Quetiapine *(Seroquel,* *Seroquel XR)*	400-800 mg/day, divided BID or XR QHS	**SIDE EFFECTS** Somnolence, orthostasis, possible ocular effects Weight gain, ↑ lipids, ↑ glucose Low EPS risk – often used for psychosis in Parkinson disease QT prolongation (lower risk) **NOTES** Take XR at night, without food or with a light meal (≤ 300 kcal)
Risperidone *(Risperdal, Risperdal Consta,* *Risperdone M-TAB ODT, Perseris)* Also approved for irritability associated with autism	4-16 mg/day, divided *Risperdal Consta*: IM injection, give every 2 weeks, 25-50 mg *Perseris*: SC injection, give monthly	**SIDE EFFECTS** Somnolence EPS, especially at higher doses ↑ prolactin – sexual dysfunction, galactorrhea, irregular/missed periods Orthostasis Weight gain, ↑ lipids, ↑ glucose QT prolongation **NOTES** > 6 mg: ↑ prolactin and ↑ EPS

PSYCHOSIS IN PARKINSON DISEASE

Pimavanserin is an inverse agonist and antagonist at serotonin 5-HT2A receptors and to a lesser extent at serotonin 5-HT2C receptors. It is approved for treatment of hallucinations and delusions associated with Parkinson disease psychosis. It does not affect dopamine receptors, and does not worsen motor symptoms of Parkinson disease.

DOSE- 34 mg PO daily (two 17 mg tablets)

SIDE EFFECTS Peripheral edema,confusion

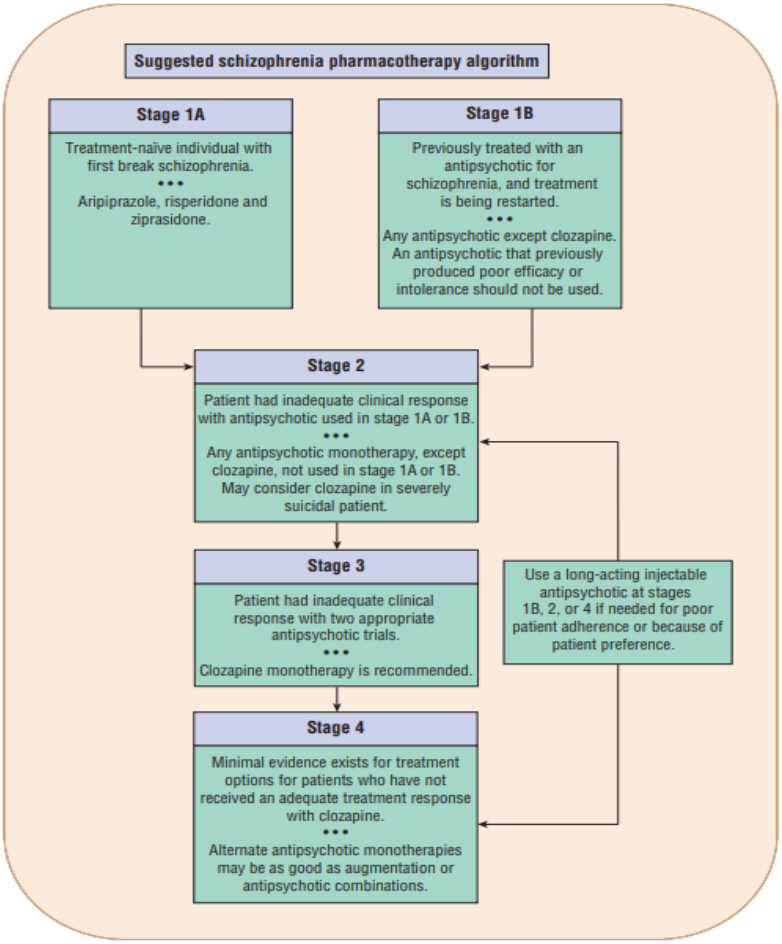

FIGURE 84-1 Suggested pharmacotherapy algorithm for treatment of schizophrenia. Schizophrenia should be treated in the context of an interprofessional model that addresses the psychosocial needs of the patient, necessary psychiatric pharmacotherapy, psychiatric co-occurring mental disorders, treatment adherence, and any medical problems the patient may have. See the text for a description of the algorithm stages. .
Data from References 26–30.

COMPLICATIONS-

TARDIVE DYSKINESIA

Tardive dyskinesia (TD) is a complication that can occur with dopamine receptor blockade, as with antipsychotics. TD can cause irreversible symptoms that include uncontrollable movements in the tongue, face, trunk and extremities and can interfere with walking, talking and breathing

DRUG	DOSING	SAFETY/SIDE EFFECTS/MONITORING
Valbenazine *(Ingrezza)*	Start 40 mg PO daily, increase in 1 week to 80 mg PO daily Moderate-severe hepatic impairment: adjustment required CYP2D6 poor metabolizer: consider dose reduction	**WARNINGS** Somnolence, QT prolongation (avoid in long QT syndrome)
Deutetrabenazine *(Austedo)* Also approved for chorea associated with Huntington's disease	Start 6mg PO BID, increase weekly based on response (max 48 mg/day) Concurrent strong CYP2D6 inhibitors or CYP2D6 poor metabolizer: max dose 36 mg/day	**CONTRAINDICATIONS** Hepatic impairment, administration with tetrabenazine or valbenazine, administration with an MAO inhibitor (within 14 days) **WARNINGS** Somnolence, QT prolongation (avoid in long QT syndrome)

.Valbenazine wasthe first medication approved fortreatment of TD.It reversibly inhibits vesicular monoamine transporter 2 (VMAT2), a transporter that regulates monoamine uptake from the cytoplasm to the synaptic vesicle forstorage and release.

Deutetrabenazine, is another VMAT2 inhibitor that is approved for TD. This drug is also used to treat chorea associated with Huntington's disease

EXTRAPYRAMIDAL SIDE EFFECTS (EPS)

Extrapyramidal side effects (EPS) are a group of side effects related to irregular movements.

Dystonias:prolonged contraction of muscles during drug initiation, including painful muscle spasms; life-threatening if airway is compromised. Higher risk with younger males. Centrally-acting anticholinergics (diphenhydramine,benztropine) can be used for prophylaxis or treatment.

Intramuscular or IV anticholinergics are the treatment of choice for dystonias, with benzodiazepines being a second-line option. Benztropine 2 mg or diphenhydramine 50 mg can be given IM or IV, whereas diazepam 5 to 10 mg by slow IV push or lorazepam 1 to 2 mg intramuscularly are treatment alternatives. Relief from the dystonia is typically seen within 15 to 20 minutes of an intramuscular injection or within 5 minutes of IV administration

Akathisia: restlessness with anxiety and an inability to remain still; treated with anticholinergics (diphenhydramine, benztropine),benzodiazepines or propranolol

Parkinsonism: looks similar to Parkinson disease, with tremors, abnormal gait and bradykinesia; treat with anticholinergics, or propranolol if tremor is the main symptom.

Tardive dyskinesias (TD): abnormal facial movements, primarily in the tongue or mouth; higher risk with elderly females. TD can be irreversible. Must stop the drug and replace with an SGA with low EPS risk (e.g., quetiapine, clozapine). This occurs due to off regulation of dopamine

Dyskinesias: abnormal movements; more common with dopamine replacement for Parkinson disease.

TABLE 84-8 Agents Used to Treat Extrapyramidal Side Effects

Generic Name	Equivalent Dose (mg)	Daily Dosage Range (mg)
Antimuscarinics		
Benztropine[a]	1	1-8[b]
Biperiden[a]	2	2-8
Trihexyphenidyl	2	2-15
Antihistaminic		
Diphenhydramine[a]	50	50-400
Dopamine Agonist		
Amantadine	NA	100-400
Benzodiazepines		
Lorazepam[a]	NA	1-8
Diazepam	NA	2-20
Clonazepam	NA	2-8
β-Blockers		
Propranolol	NA	20-160

NA, Not applicable.

[a]Injectable dosage form can be given intramuscularly for relief of acute dystonia.

[b]In treatment-refractory cases, dosage can be titrated to 12 mg/day with careful monitoring; nonlinear pharmacokinetics have been reported.

NEUROLEPTIC MALIGNANT SYNDROME

Antipsychotics used to be called neuroleptics. Neuroleptic malignant syndrome (NMS) is rare but is highly lethal. It occurs most commonly with the FGAs and is due to D2 blockade

Signs Include

- Hyperthermia (high fever, with profuse sweating)
- Extreme muscle rigidity (called "lead pipe" rigidity), which can lead to respiratory failure
- Mental status changes
- Othersigns can include tachycardia, tachypnea and blood pressure changes

Laboratory Results

- T creatine phosphokinase and t white blood cells

Treatment

- Taper off the antipsychotic quickly and consider another choice (quetiapine or clozapine).
- Provide supportive care: cardiorespiratory and hemodynamic support and control of electrolyte balance.
- Cool the patient down: cooling bed, antipyretics, cooled IV fluids.
- Muscle relaxation with benzodiazepines or dantrolene (Ryanodex, Dantrium, Revonto) is sometimes used, and some cases may require a dopamine agonistsuch as bromocriptine

Explain the symptoms and pharmacotherapy for schizophrenia (5)

Write treatment algorithm of schizophrenia (5)

How atypical antipsychosi differ from typical antipsychotic(2)

DOPAMINE HYPOTHESIS

The dopamine hypothesis of schizophrenia postulates that hyperactivity of dopamine D2 receptor neurotransmission in subcortical and limbic brain regions contributes to positive symptoms of schizophrenia, whereas negative and cognitive symptoms of the disorder can be attributed to hypofunctionality of dopamine D1 receptor neurotransmission in the prefrontal cortex

PICKING THE BEST ANTIPSYCHOTIC

Each patient will respond differently to antipsychotics. A trial of different drugs may be necessary before finding the best fit.

All antipsychotics carry some risk of the following side effects. On a case, evaluate patient-specific factors to help pick the best drug and avoid drugs that could cause harm.
FGA: ↑ EPS, ↓ weight gain/metabolic side effects
SGA: ↓ EPS, but ↑ metabolic effects

EPS
Lowest risk – Quetiapine (recommended in patients with Parkinson disease who require antipsychotics)
Highest risk - FGAs, risperidone, paliperidone

Metabolic side effects
Highest risk – Clozapine, olanzapine, quetiapine
Moderate risk – Risperidone, paliperidone
Lower risk – FGAs, aripiprazole, ziprasidone, lurasidone and asenapine

QT prolongation
Highest risk – Thioridazine, haloperidol, ziprasidone

Hematological effects
Highest risk – Clozapine (agranulocytosis)

Cerebrovascular events (e.g., TIA, stroke)
Highest risk – Risperidone

↑ Prolactin
Highest risk – Risperidone, paliperidone

Seizure
Highest risk – Clozapine (dose-dependent)

Aute IM are often mixed with other drugs (called “cocktails"), such as benzodiazepines for anxiolytic and sedative effects, and anticholinergics to reduce dystonic risk (e.g., the “Haldol cocktail" contains haloperidol, lorazepam and diphenhydramine).